Design Optimization of a Mars Rover's Rocker-Bogie Mechanism using Genetic Algorithms

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Summary: In this work we present our findings in the optimization of a particular suspension system known as the "rocker-bogie" for our Mars Rover. This type of mechanism has been used on most of the rovers on Mars and has proved to be a simple and elegant design. This work defines the most important performance metrics for a planetary rover. A Genetic Algorithm is implemented and used to optimize the geometry and kinematics of the rover's wheel suspension system subject to the defined performance metrics. A Genetic Algorithm is chosen for optimization due to its simplicity of implementation and fast convergence to an optimum within a complex and unknown search space.

This work shows the effectiveness of the optimization of a rocker-bogie suspension system using a Genetic Algorithm. It also reveals that the resulting system meets all constraints and that significantly reduces the error of individual performance metrics and the overall system. It is shown that the overall fitness of the rover suspension system can be increased by an average of 28% after 100 iterations compared to an initial guess. All performance metrics defined are improved significantly throughout the optimization.

The method can be applied to different types of rovers in order to optimize the wheel suspension mechanism's geometry.

Keywords: Mars Rover, Rocker-Bogie Mechanism, Design Optimization, Genetic Algorithm

Introduction and Background

The type of locomotion used by a mobile robot is crucial for the robot to perform its task and reach its goal in a given environment. This work focuses on the optimization of the design of a planetary rover's wheel suspension system subject to optimizing well defined mobility metrics.

As robots evolve from industrial fixed base robots to autonomous mobile platforms, the concept of locomotion in robotics becomes much more important. Similar to nature, also robot locomotion must be adapted to the given terrain or task. The optimal type of locomotion must be applied in a challenging environment.

The scope of this work is to design an efficient Mars Rover suspension system and to develop and implement a Genetic Algorithm methodology which optimizes the design of the locomotion system and can be applied to diverse mechanisms or other problems.

As the Mars Rover is a mobile robot, the wheel suspension system of the rover is most crucial. It allows for movement, mobility and stability of the robot while it is travelling through a Mars environment. The rover must be able to traverse over obstacles of at least half its wheel diameter and keep its stability on slopes or other rough or hazardous terrain. Hence, the

suspension system is to be optimized using a Genetic Algorithm.

A planetary rover is a mobile robot which is especially designed to move on a planet surface. Early rovers were teleoperated like the Lunokhod I [1] while recent ones are fully autonomous, such as FIDO [2]. The rover has to be very robust and reliable, as it has to withstand dust, strong winds, corrosion and large temperature changes. Most rovers are powered by batteries which are recharged by solar panels during the day. Therefore, the rover must position itself in such a way that the solar energy received is maximized. The rover's locomotion system is crucial to enable it to reach a goal, conduct experiments, gather data and to position itself. Generally three main types of rover locomotion exist, namely wheeled, legged and caterpillar locomotion [3], [4]. The key difference between the miscellaneous designs of planetary robots lies in the type of locomotion system. Even though many legged and hybrid robots have been presented in literature, most researchers still focus on wheeled locomotion for rovers. The favoured design for a wheeled planetary rover's suspension system is the rocker-bogie mechanism. Numerous variations of the this mechanism have been presented in literature. The FIDO rover [2] and the Sojourner [5] have six independently steered and driven wheels suspended from a rocker-bogie mechanism. The Rocky7 Rover [6] has a similar suspension system, but only the front wheels are steered. The Nanorover [7] and the Nomad [8] have four steered wheels suspended from two bogies, a variation of the rocker-bogie mechanism. The CRAB Rover [9] uses two parallel bogie mechanisms on each side to overcome obstacles.

Lamon et al. [10] optimize a simplified quasi-static model of the six-wheeled rover Shrimp. The motor torques are optimized subject to minimizing the slip of the wheels, hence the odometric error and power consumption are minimized. The optimization seeks for an optimum in the constrained solution space given an initial solution. Li et al. [11] derive a mathematical model to optimize rover suspension parameters which define the geometry of the rocker-bogie. The objective is to minimize the energy consumption, the vertical displacement of the rover's centre of mass and its pitch angle. The authors make use of a sequential quadratic programming algorithm. Iagnemma and Dubowsky [12] introduce an algorithm which optimizes individual wheel ground contact forces. The objective is to optimize traction and to minimize power consumption.

Rocker-Bogie Modelling

The Rocker-Bogie Mechanism

The rocker-bogie suspension system was first used for the Mars Rover Sojourner and is currently NASA's favoured design for rover wheel suspension. The intelligently designed wheel suspension allows the vehicle to traverse over very uneven or rough terrain and even climb over obstacles. The rocker-bogie allows the chassis of the rover to average its pitch over all wheel deflections, while still maintaining load equalization on all wheels and avoiding a low oscillation frequency.

The suspension has symmetric structure on both sides of the rover chassis and supports a total of six wheels, three on each side. The mechanism is constructed from two rigid parts. The link connected with a rotary joint to the chassis is called the *rocker*. The *bogie* is connected to the rocker via another rotary joint, as shown in the kinematics diagram in Fig. 1. The rockers are connected to each other via a differential mechanism, which allows the rover body to minimize its pitch.

The rocker-bogie suspension mechanism has two main advantages. The total load that is put

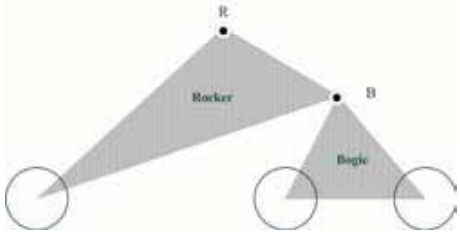

Fig. 1: Kinematics Diagram of a Rocker-Bogie Mechanism

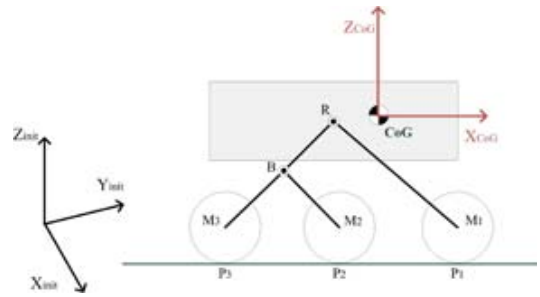

Fig. 2: Points M_i, P_i, B, R and Reference Frames and their Locations on the Rocker-Bogie

on the rover is distributed evenly over all wheels. This ensures equal working conditions for all wheels and prevents from excessive sinkage of wheels in soft terrain [13]. The second advantage is that all wheels stay in touch with the ground while climbing obstacles. This ensures stability and prevents the rover from tilting on a slope. The rocker-bogie is a passive suspension system and has no axles or springs, all wheels are kept in contact with the ground without any actuators, decreasing power consumption of the rover.
Each of the six wheels is independently driven and steered. This also allows the rover to turn in place or move sidewards. The system therefore is redundant and can move forwards, backwards and sidewards.

Simplified System Model

The mathematical model basically takes into account the entire geometry of the rocker-bogie, in terms of link lengths and coordinates of connection points. Instead of concentrating on the exact geometry, it is important to make use of all connection points between links. Therefore, a L-shaped link can be constructed in another way in the realistic model as long the end points are kept. The exact shape of the links is then to be designed according to the general skeleton, which is subject of the latter geometry optimization.
In the optimization process the coordinates of the wheel midpoints (M_i), the coordinates of the wheel terrain contact points (P_i) and the coordinates of both rocker (R) and bogie (B) connection points are used, where $i = \{1 \ldots 6\}$ is the number of wheels. All these coordinates are computed with reference to a global frame ($X_{init}, Y_{init}, Z_{init}$) and the body-fixed frame ($X_{CoG}, Y_{CoG}, Z_{CoG}$), located at the centre of gravity (CoG). The forward kinematics of the rover are derived and the coordinates of points R, B, P_i and M_i are found, shown in Fig. 2.

When deriving the forward kinematics of the rocker-bogie suspension system, it is assumed that the terrain and its slope are known. Also the gradient of the ground is known at every point of the terrain. This is not a realistic assumption, as in the real world it is a very challenging task to find the terrain's shape. Furthermore, a point contact between wheel and terrain is assumed. This assumption is true for a rigid wheel and a relatively firm ground.
Another assumption is that the rover does not experience sideslip. Furthermore, it is assumed there are no moments acting at the wheel-terrain interface. In this work, all links are assumed to be rigid bodies. The maximum travelling speed for the rover is assumed to be $0.25\ \frac{m}{s}$,

the maximum acceleration is assumed to be $1\ \frac{m}{s^2}$. As maximum velocity and acceleration are relatively small, the rover can be modeled as a quasi-static system and all dynamic effects are neglected [10]–[12].

Performance Evaluation and Metrics

Performance evaluation of a system is a necessary process, which provides many information about the system and helps to improve future systems. Prior to an optimization of the rocker-bogie suspension system, it is important to define the performance metrics. The performance metrics form the basis for the objective function, which is to be maximized during the optimization process.

Wheel-Terrain Contact Forces

The analysis of the rover's suspension system's design demands for the knowledge of the wheel-terrain contact interaction. It is assumed, that each rigid wheel makes contact with the known and relatively firm ground at a single point, P_i. The vectors from the contact points P_i to the rover's centre of gravity are given by $\mathbf{p_i} = [p_i^x, p_i^y, p_i^z]^T$ as illustrated in Fig. 3. The contact force vector $\mathbf{f_i} = [T_i, N_i, S_i]^T$ where T_i is the traction force at wheel-terrain contact point P_i, N_i is the normal force and S_i is the force acting sidewards. The vectors $\mathbf{p_i}$ and $\mathbf{f_i}$ are expressed in the rover's bodyframe.
The 6×1 vector $\mathbf{f_s}$ is expressed in the rover's bodyframe and is located at the rover's centre of gravity. It represents all forces and moments acting on the rover body, such as gravitational forces, inertial forces, manipulation and driving forces. The decomposition of $\mathbf{f_s}$ is shown in Fig. 4, where α_i is the angle of incline of the ground at the wheel-terrain contact point P_i. As suggested by Iagnemma and Dubowsky [12] a set of quasi-static force balance equations are written as

$$\begin{bmatrix} \mathbf{I} & \cdots & \mathbf{I} \\ \begin{matrix} 0 & p_1^z & p_1^y \\ -p_1^z & 0 & -p_1^x \\ -p_1^y & p_1^x & 0 \end{matrix} & \begin{matrix} \cdots \\ \cdots \\ \cdots \end{matrix} & \begin{matrix} 0 & p_6^z & p_6^y \\ -p_6^z & 0 & -p_6^x \\ -p_6^y & p_6^x & 0 \end{matrix} \end{bmatrix} \begin{bmatrix} \mathbf{f_1} \\ \vdots \\ \mathbf{f_6} \end{bmatrix} = \mathbf{f_s} \tag{1}$$

in order to find the wheel-ground interaction forces. In (1) $\mathbf{I}$ is the 3×3 identity matrix and all dynamic effects are neglected.
The matrix equation (1) can be rewritten as $\mathbf{Gx} = \mathbf{f_s}$, where $\mathbf{G}$ is a function of the rover's geometry, the wheel-terrain contact location and the wheel-terrain contact angles α_i. The vector $\mathbf{f_s}$ is the input vector of all forces acting on the rover. The equation is solved for $\mathbf{x} = [T_1, N_1, S_1 \ldots T_n, N_n, S_n]^T$, $n = 6$, using the Moore-Penrose pseudo inverse, defined as $\mathbf{x}^+ = (\mathbf{G}^T\mathbf{G})^{-1}\mathbf{G}^T\mathbf{f_s}$. For this calculation matrix $\mathbf{G}$ is assumed to have full rank. This approach is very inexpensive, however, it might not satisfy the system's physical constraints.

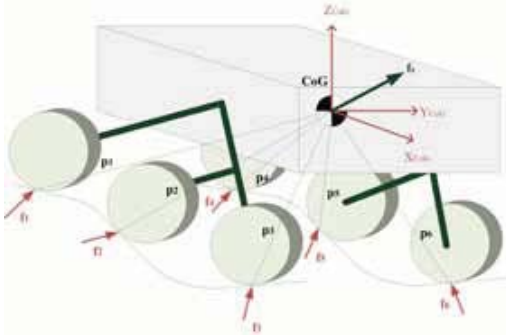

Fig. 3: Rover with six Wheels on Rough Terrain

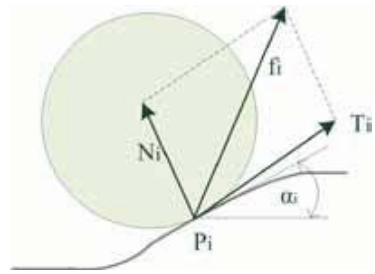

Fig. 4: Derivation of Forces at Wheel-Terrain Contact Point

Power Consumption

As a planetary rover can not recharge via a power plug or change batteries during its mission, it is very important that power consumption of the rover is low. Power consumption is mainly the energy being used by the driving and steering motors. As suggested by [12], it is given by $P = \frac{R_M g_M^2}{K_t^2} \sum_{i=1}^{n} \Gamma_i^2$, where R_M is the motor resistance, g_M is the motor gear ratio, K_t is the motor torque constant and Γ_i is the torque applied to the *i*th motor. As the applied torque is the product of traction force T_i at wheel *i* and the same wheel's radius r, power consumption can be written as $P = \frac{R_M g_M^2 r^2}{K_t^2} \sum_{i=1}^{n} T_i^2$. It is obvious that power consumption is proportional to the sum of all squared traction forces T_i. In order to minimize the power consumption of the rover during travel, the sum of the squared traction forces is to be minimized.

Traction and Slip

The rover must maintain good wheel traction in challenging rough terrains. If traction is too high, the vehicle consumes a lot of power in order to overcome the force and move. If traction is too low, the rover is not able to climb over obstacles or inclined surfaces. Slip occurs when the traction force at a wheel-terrain contact point is larger than the product of the normal force at the same wheel and the friction coefficient. Hence, no slip occurs if the condition $T_i \leq \mu N_i$ is satisfied. In reality it is very challenging to determine the precise friction coefficient μ for the interaction of two surfaces. As an approximation of the friction coefficient, a virtual friction coefficient μ^* is computed, according to Lamon et al. [10], as the ratio between traction and normal force at one wheel-terrain contact point.

$$\mu^* = \max_i \{\frac{T_i}{N_i}\} \tag{2}$$

The parameter μ^* is similar to a friction coefficient. If the constraint $\mu^* \leq \mu$ is satisfied, no slip occurs. Hence, μ^* is targeted to be minimized. According to [10] and [14] the optimum solution is found if μ^* is equal for all n wheels.

Load Equalization

Load equalization over all n wheels ensures equal work conditions for all wheels. The following Fig. 5 shows the two dimensional mechanics model of the rocker-bogie suspension. Weights G, normal forces N and traction forces T are shown as well as geometric dimensions. The weight G_b is the total weight of the rover body while G_i denotes the respective weight from the rocker-bogie mechanism acting on wheel i. A quasi-static force balance is conducted, as suggested by Li et al. [11], and the conditions $x_B = \frac{3}{2}x_C$ and $x_B = \frac{1}{2}(x_1 + x_2)$ are derived for load equalization. These geometrical constraints for the suspension structure are used to maximize load equalization.

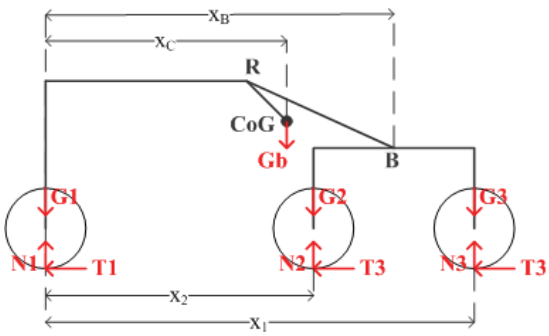

Fig. 5: Load Equalization over all n Wheels

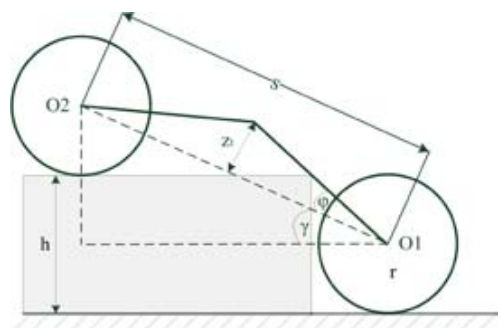

Fig. 6: Geometric Trafficability of the Bogie

Geometric Trafficability

Geometric trafficability ensures that the suspension structure does not interfere with the ground. The analysis of geometric trafficability is only done for the bogie, as the swinging scope for the bogie is much bigger than that of the rocker. The analysis of geometric trafficability is done according to [11]. From geometry the maximum vertical distance z_t between the centre of the wheels to the edge of the obstacle is determined according to (3), where r is the wheel radius and h and s are geometric properties, seen in Fig. 6.

$$z_t = \frac{s}{\sqrt{\left(\frac{r\sqrt{s^2-h^2}+(h-r)h}{(h-r)\sqrt{s^2-h^s}-hr}\right)^2 + 1}} \tag{3}$$

$$z_t \leq c \tag{4}$$

From geometry the maximum vertical distance z_t between the centre of the wheels to the edge of the obstacle is determined. Geometric trafficability is given if the parameter z_t is larger or equal to the rover's ground clearance.

Lateral and Longitudinal Stability

The rover is said to be stable when it is in a quasi-static state in which it does not tilt over. The simplest approach to find the static stability is using the geometric model, which

is commonly referred to as stability margin. As the asymmetric suspension system of the passively articulated rover has a great influence on the vehicle's effective stability, a more advanced approach is using a static model. The geometric a) and the statical b) model for the rocker-bogie suspension system are illustrated in Fig. 8.

The lateral stability of the rover ensures that the rover does not tip sideways. As the rover has

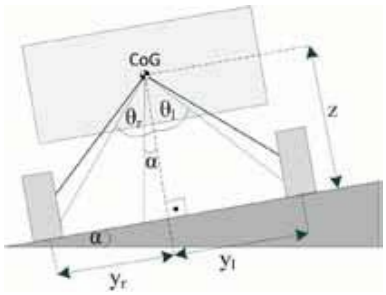

Fig. 7: Lateral Stability of the Rover

Fig. 8: Stability Models for Evaluation

two symmetric sides, the geometric model is used to find the lateral stability of the vehicle, as suggested by Thüer [14]. Lateral stability is computed by finding the minimum allowed angle on the slope before the rover tips over. Lateral stability is ensured if this angle is smaller than the maximum angle of incline α on the slope at the wheel-terrain contact points. The angles θ_l and θ_r are obtained geometrically. The overall stability angle θ_{stab} can be computed by

$$\theta_{stab} = \min\left(\theta_{\mathrm{r}}, \theta_{\mathrm{l}}\right) \tag{5}$$

Lateral stability of the rover is ensured if the overall stability angle $\theta_{stab} \geq \alpha$.

The computation of the longitudinal stability of the rover makes use of a statical model as it is not symmetric in longitudinal direction. Using a statical model, the mechanical properties of the suspension system are taken into account. According to [14], longitudinal stability of the vehicle is given when all wheels have ground contact and the condition $N_i > 0$ is satisfied, where N_i is the normal force at wheel i. It should be noted that even though this condition is compulsory for the statical model to work, a physical rover does not necessarily tip if a wheel looses contact to the ground. However, it is less steerable.

Sinkage

A rover traveling over terrain is subject to sinkage, depending on the properties of the soil as well as loading, geometry and shape of the wheels. Apostolopoulos [15] suggests an experimentally derived formula for the maximum sinkage experienced by a rigid wheel in weak soil. The soil is defined by soil parameters which have been tabulated by Wong [16] for various soils. An optimization of the vehicle's geometry seeks to reduce the maximum sinkage of the wheels on soil.

Pitch Variation

As the rover is powered by solar energy, its solar panels should be perpendicular to the sun rays at all times. Therefore, very low variation in the rover's pitch angle is desirable.

A differential mechanism connects the two rocker-bogies. As the rover traverses over rough terrain its pitch is the average of the two rocker deflections. By choosing an appropriate gear ratio the overall pitch of the rover is optimized.

Design Optimization of the Rocker-Bogie

This work uses a Genetic Algorithm in order to find an optimal solution for the rocker-bogie suspension. The Genetic Algorithm shows some advantages over other optimization techniques. It optimizes with discrete or continuous variables and does not require derivative information. Also, it optimizes a large number of variables with extremely complex search spaces and is therefore a global search algorithm.
The Genetic Algorithm is started with an initial population, which consists of a number of chromosomes, each representing a possible solution to the optimization problem. The aim of the optimization is to minimize the cost and hence maximizing the *fitness*. Each chromosome is associated with a certain fitness, given by the fitness criterion or objective function. In order for the population to evolve, in each generation *selection* of the parent chromosomes takes place. These parents then pair up to produce new offspring that replaces the old population. Different methods for selecting parents are random selection, roulette wheel selection, ranked selection and tournament selection. Discarding of all chromosomes in each new generation might not be very efficient, when searching for the best solution. Therefore, the principle of *elitism* is introduced. This means that the fittest chromosome of each population is kept. Hence, the best solution of the last population of chromosomes is copied to the new population and is ensured to survive. The process of *mating* is the creation of new offspring from the parents, selected before. Every two parents produce one new offspring by the means of *crossover*. The first parent passes on some properties while the second parent passes on others. The new offspring consequently has parts of both parents. It is inserted into the new population and might be chosen to be a parent chromosome in the next generation, according to its fitness. *Mutation* is a way to explore the search space. While mating is limited by those chromosomes in the population, mutation can introduce new traits in the population. These new attributes may increase a population's fitness or may decrease it. Increasing the number of mutations distracts the algorithm from converging to one popular solution, and hence makes the Genetic Algorithm a global search algorithm. The generational process is terminated when the termination criterion is reached.

During the rocker-bogie optimization the geometry of the simplified model is to be optimized subject to the objective function, shown in equation (6)

$$\begin{aligned} \text{fitness} &= -w_1 s\mu^* - w_2\varepsilon_{\mu^*} + w_3\left(s-1\right)P + w_4 c_{lat} + w_5 c_{long} \\ &\quad -w_6\left(\varepsilon_1+\varepsilon_2\right) + w_7 c_{traff} - w_8 z_{rw} - w_9\theta_{rover} \end{aligned} \tag{6}$$

where w_1 to w_9 are the weights, as shown in Table 1. Parameter μ^* is the virtual friction coefficient. By minimizing the error ε_{μ^*} the difference of virtual friction coefficient at each

Table 1: Weights for Performance Metrics during Optimization Series

Weight no.	w_1	w_2	w_3	w_4	w_5	w_6	w_7	w_8	w_9
Weight	2	1	2	1	5	3	2	1	1

Table 2: The Upper (UL) and Lower Limits (LL) of the Optimization Variables

Variable	x_R	y_R	z_R	γ_{RB}	x_{B1}	y_{B1}	x_{B2}	y_{B2}	c	L_{RB}	j
LL $[mm]$	100	100	100	$90[deg]$	100	100	100	100	20	50	$1[-]$
UL $[mm]$	500	300	200	$180[deg]$	200	300	200	300	100	500	$5[-]$

wheel is decreased, leading to similar work conditions for all wheels. Parameter P is proportional to the power consumption. In rough terrain, traction should be maximized, while in benign terrain, power consumption should be minimized. The switching function s ensures this property and is given by

$$s = \begin{cases} 1, & \text{if} \max(|\alpha|) > C \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where α_i is the terrain's angle of incline at each wheel-terrain contact point. C is an arbitrary threshold, which defines what is considered rough or benign terrain. Furthermore, c_{lat}, c_{long} and c_{traff} are parameters, which are large $(= 1000)$ if lateral stability, longitudinal stability and trafficability are given, respectively, and zero otherwise. Parameters ε_1 and ε_2 are to be minimized, in order to ensure load equalization. The sinkage z_{rw} and the rover pitch θ_{rover} are minimized.
Geometrical parameters of the rocker-bogie model as well as the gear ratio j of the differential mechanism connecting both rocker-bogies are to be optimized and represented in the chromosome, as illustrated in (8) and shown in Fig. 9.

$$\text{chromosome} = [\ x_R \ \ y_R \ \ z_R \ \ \gamma_{RB} \ \ x_{B1} \ \ y_{B1} \ \ x_{B2} \ \ y_{B2} \ \ c \ \ L_{RB} \ \ j\ ] \tag{8}$$

All geometrical parameters are constrained by real numbers representing the physical constraints, these are shown in Table 2.

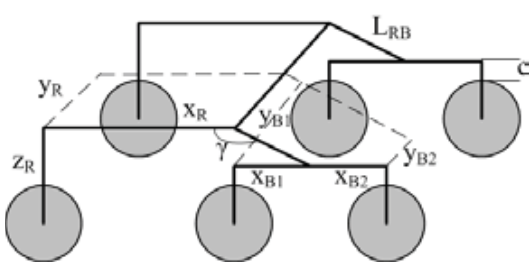

Fig. 9: Geometry Model of Rocker-Bogie showing all Geometrical Parameters

During optimization the ground profile assumed is a hill with specified angle of inclination α, but can be changed by the user.
The Genetic Algorithm evaluates the fitness of each chromosome in the population and therefore quantifies the goodness of each of these specific combinations of link lengths and angles of the rocker-bogie mechanism. It then mates two good solutions in order to find even better combinations of variable values. The fitness is determined by the objective function, which includes all the performance metrics described above. The traction is optimized while the slip and power consumption of the rover are minimized. Furthermore, load equalization is maximized and sinkage of the wheels and the variation of the rover's pitch are minimized. Also, geometric trafficability, lateral and longitudinal stability are to be ensured by minimizing the objective function and hence maximizing the fitness of the best solution, represented by the best chromosome in the final population. All performance metrics are weighted in the objective function. By adjusting the weights, the user can emphasize one metric's importance over another one's. By maximizing the fitness function, the performance of the rover is enhanced. As fitness is a function of the rover suspension system geometry, the optimum for the geometrical parameters can be found.

Results

The principles of elitism and mutation are applied with a mutation rate of 0.2. The selection type chosen for this optimization is tournament selection. The rigid wheels have a diameter of $d_w = 170\ mm$ and a width of $b_w = 75\ mm$. The rover is moving over sand and the maximum obstacle height to be climbed is $h = 170\ mm$, equal to the wheel diameter. The Genetic Algorithm is terminated after 100 generations. The optimization algorithm is run 25 times during the series. Fig. 10 shows the progression of the best fitness within one population, representing 70 chromosomes, for 100 generations. Fig. 11 visualizes the progression of the average fitness within the same population. All performance metrics are weighted as shown in Table 1. Both, Fig. 10 and 11, show an example regression which is similar for all 25 runs. The values found for the optimization parameters are different for every run. No two solutions found are exactly the same. Nonetheless, most values are in a similar range and show a specific trend for the particular dimension.

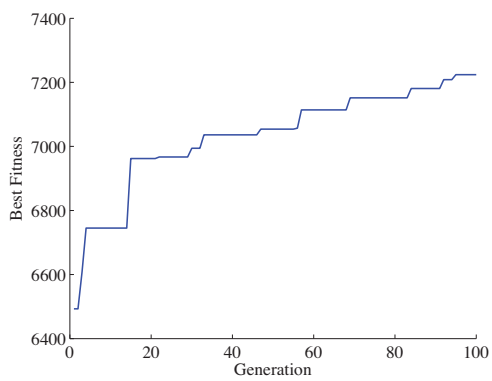

Fig. 10: Progression of the Best Fitness during Optimization over 100 Generations

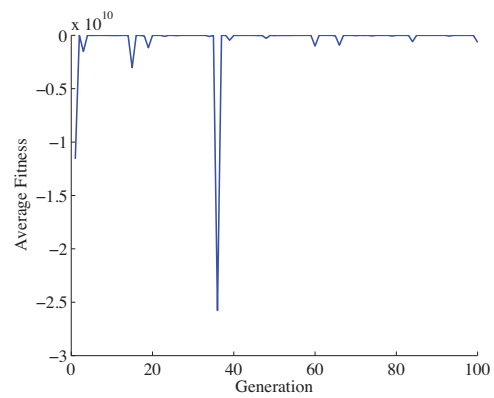

Fig. 11: Progression of the Average Fitness during Optimization over 100 Generations

Analysis and Discussion

In general, it is shown that the Genetic Algorithm significantly increases the fitness of the chromosome over 100 generations and therefore optimizes the geometry of the rover's rocker-bogie suspension system subject to the performance metrics described.

Fig. 10 illustrates the best fitness within the evolving population, which represents a set of 70 different solutions. It is clear, that the fitness is increased stepwise. The shape of the progressing fitness is given due to the principle of elitism. The best fitness is that of the best solution which is kept in the evolving population until a better solution with a higher fitness is found. So the solution with the highest fitness is exploited, as the value for best fitness does not decrease during the optimization. The results show a large increase of the best fitness within the population of possible solutions between generation one and 100. While absolute increase of the fitness is 1527 on average compared to a random guess within the physical constraints, the relative increase is between 7% and as much as 104%, with an average of 28% among the 25 runs. The optimization algorithm improves the fitness significantly over 100 iterations.

Fig. 11 shows the progression of the mean fitness within one population. It is obvious that many areas of the search space are explored as the mean value for the fitnesses oscillates over a wide range. The principle of mutation makes the optimization a global search within a large search space. As point mutations are inserted into a new population, other parts of the search space are explored.

The optimal solution found during each run is different to the others, but the optimized values found for each parameter, represented by the chromosome, are in a similar range and show a trend for each optimization parameter. This suggests that multiple designs for the rocker-bogie show similar values for the fitness and are equally good in terms of optimizing the performance metrics. Hence, the dimensions of the suspension system, such as link lengths and angles between links, are strongly interdependent. In general, the rocker-bogie mechanism is found to be optimal for certain structural details. The total width of the rover is found to be optimal if it is large, while total height should be small. Both of these details can be explained by the desire for lateral and longitudinal stability of the rover. The connection point between rocker and rover body is not located in the centre but to the front of the rover. The angle between rocker and bogie is large, while the connecting link is rather short. This ensures a long vehicle but less weight than for a long connecting link. The middle and rear wheel are not in line with the front wheel for an optimal solution but are located further from the rovers centre. This detail also increases stability, especially during traversing over an obstacle. Furthermore, the ground clearance is kept as small as possible while still ensuring geometric trafficability of the rover. After optimization of the total fitness of a solution, each performance metric was computed separately using the optimal values for the rocker-bogie geometry found by the Genetic Algorithm. Validating each individual performance metric shows that the quality of every metric was increased during optimization by a large percentage. Furthermore, lateral stability, longitudinal stability and geometric trafficability are given in 100% of the cases. The results show that not only the total quality of the rocker-bogie mechanism was increased during the optimization, but also each performance metric could be optimized using a Genetic Algorithm.

Conclusion

This work describes the modelling of a quasi-static rocker-bogie suspension system for a Mars Rover, followed by the identification of several performance metrics which are to be optimized if the rover is to execute its tasks and reach its goals in the best possible way. Subsequently, an optimization method is implemented. The applied method is a Genetic Algorithm which is an evolutionary computing method.

Several performance metrics have been introduced and described which give a thorough basis for the evaluation of the rover's rocker-bogie suspension system. Furthermore, a Genetic Algorithm is described and applied to the optimization problem. The results of the optimization and the analysis of the same shows that the rover's suspension mechanism's geometry is optimized. After optimization it shows a better performance and its abilities to traverse are improved. Overall performance is increased by an average of 28% as well as the individual performance metrics.

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References

[1] Cherkasov, II and Shvarev, VV, "Soviet investigations of the mechanics of lunar soils", *Soil Mechanics and Foundation Engineering*, Vol.10, No. 4, 1973, pp. 252-256.

[2] Schenker, PS and others, "FIDO: A field integrated design & operations rover for Mars surface exploration", *i-SAIRAS, Montreal, Canada*, 2001.

[3] Siegwart, R. and Lamon, P. and Estier, T. and Lauria, M. and Piguet, R., "Innovative design for wheeled locomotion in rough terrain", *Robotics and Autonomous systems*, Vol. 40, No. 2-3, 2002, pp. 151-162.

[4] Estier, T. and Crausaz, Y. and Merminod, B. and Lauria, M. and Piguet, R. and Siegwart, R., "An innovative space rover with extended climbing abilities", *Proceedings of Space and Robotics*, 2000, pp. 201-206.

[5] Mars Pathfinder Rover by NASA, http://nssdc.gsfc.nasa.gov/nmc/spacecraftDisplay.do?id=MESURPR, 2009.

[6] Rocky 7: A next generation mars rover prototype, "Rocky 7: A next generation mars rover prototype", *Journal of Advanced Robotics*, 1997.

[7] Mars Pathfinder Rover by National Aeronautics and Space Administration (NASA), http://www-robotics.jpl.nasa.gov/tasks/showTask.cfm?FuseAction=ShowTask&TaskID=109&tdaID=888881, 2009.

[8] Wettergreen, D. and Bapna, D. and Maimone, M. and Thomas, G., "Developing nomad for robotic exploration of the atacama desert", *Robotics and Autonomous Systems*, Vol. 26, No. 2-3, 1999, pp. 127-148.

[9] Thüer, T. and Lamon, P. and Krebs, A. and Siegwart, R., "CRAB-Exploration rover with advanced obstacle negotiation capabilities", *Proceedings of the 9th ESA Workshop on Advanced Space Technologies for Robotics and Automation*, 2006.

[10] Lamon, P. and Krebs, A. and Lauria, M. and Shooter, S. and Siegwart, R., "Wheel torque control for a rough terrain rover", *IEEE International Conference on Robotics and Automation*, Vol. 5, 2004, pp. 4682-4687.

[11] Li, S. and Gao, H. and Deng, Z., "Mobility performance evaluation of lunar rover and optimization of rocker-bogie suspension parameters", *Systems and Control in Aerospace and Astronautics, 2008. ISSCAA 2008.*, 2008, pp. 1-6.

[12] Iagnemma, K. and Dubowsky, S., "Traction control of wheeled robotic vehicles in rough terrain with application to planetary rovers", *The International Journal of Robotics Research*, Vol. 23, No. 10-11, 2004, pp. 1029.

[13] Harrington, B.D. and Voorhees, C., "The challenges of designing the rocker-bogie suspension for the Mars Exploration Rover", *37th Aerospace Mechanisms Symposium*, 2004.

[14] Thüer, T., "Mobility Evaluation of Wheeled all-terrain Robots - Metrics and Application", *PhD Thesis, Swiss Federal Institute of Technology Zurich (ETH Zurich)*, 2009.

[15] Apostolopoulos, D., "Analytical configuration of wheeled robotic locomotion", *Unpublished Ph. D. thesis, The Robotics Institute, Carnegie Melon University, Pittsburgh, PA*, 2001.

[16] Wong, J.Y., *Theory of ground vehicles*, 2001.